We have seen (Figure A2.56) that, when any two objects, A and B, interact, $F_A = -F_B$.

Using Newton's second law $F = ma$ we can write: $[ma]_A = -[ma]_B$.

Remembering that:

$$a = \frac{(v - u)}{t}$$

we can write:

$$\left[\frac{m(v-u)}{t}\right]_A = \left[\frac{m(v-u)}{t}\right]_B$$

The time of the interaction is the same for both, so: $[m(v - u)]_A = [m(v - u)]_B$.

Putting this into words: (mass × change of velocity) for A = −(mass × change of velocity) for B.

◆ **Momentum (linear), *p*** Mass times velocity, a vector quantity.

(mass × velocity) is an important concept in physics. It is called **momentum**. The momentum gained by object A = momentum lost by object B. Always. This assumes that there are no external forces.

This is covered in more detail in the next section.

TOK

The natural sciences

- What kinds of explanations do natural scientists offer?

A clockwork universe?

Everything is made of particles and it has been suggested that, if we could know everything about the present state of all the particles in a system (their positions, energies, movements and so on), then maybe we could use the laws of classical physics to predict what will happen to them in the future. The Universe would then behave like a mechanical clock. If these ideas could be expanded to include everything, then the future of the Universe would already be decided and predetermined, and the many apparently unpredictable events of everyday life and human behaviour (like you reading these words at this moment) would just be the laws of physics in disguise.

However, we now know that the laws of physics (as imagined by humans) are not always so precisely defined, nor as fully understood as physicists of earlier years may have believed. The principles of quantum physics and relativity in particular contrast with the laws of classical physics. Furthermore, in a practical sense, it is totally inconceivable that we could ever know enough about the present state of everything in the Universe in order to use that data to make detailed future predictions.

Momentum

SYLLABUS CONTENT

- Linear momentum as given by: $p = mv$ remains constant unless the system is acted upon by a resultant external force.
- Newton's second law in the form $F = ma$ assumes mass is constant whereas $F = \frac{\Delta p}{\Delta t}$ allows for situations where mass is changing.

DB linear momentum (SI unit: $\mathrm{kg\,m\,s^{-1}}$) = mass × velocity, $\boldsymbol{p = mv}$

Momentum is a vector quantity and its direction is always important.

◆ **System** The object(s) being considered (and nothing else). An **isolated system** describes a system into which matter and energy cannot flow in, or out.

The explanation at the end of the last section shows that when two objects interact, with forces between them, the change of momentum for one object is equal and opposite to the change in momentum of the other: one object gains momentum while the other object loses an equal amount of momentum. This means that the total amount of momentum is unchanged, although this is only true if no resultant external force is acting on the objects. (We describe this as an **isolated system**.). This very important principle, which is a consequence of Newton's third law, can be stated as follows:

The law of conservation of momentum: the total momentum of any system is constant, provided that there is no resultant external force acting on it.

This law of physics is always true. There are no exceptions. It is very useful in helping to predict the results of interactions like collisions. See below.

Nature of science: Models

Systems and the environment

◆ **Surroundings** Everything apart from the system that is being considered; similar to the 'environment'.

We use the term *system* to describe and limit the collection of objects we are considering. You may think of this as 'drawing a line around' an object together with all of the surrounding objects with which there are significant interactions. This is especially important when using conservation laws. Objects outside of the 'system' are usually referred to as the environment, or the **surroundings**.

In practice, any situation can be complicated and we often have to decide which objects we can ignore (assume to be outside of the system) because their effect is minimal.

Take a collision between two cars as an example. Commonly, we calculate an outcome by considering the system to be just the two cars. This will give us a quick, reasonably accurate and useful prediction for what happens immediately after impact. Such a calculation has chosen not to include the air and the road in the system. If they were included, the situation would be much more complex, but the immediate consequences of any collision may be similar.

We know that, for uniform acceleration:

$$F = ma = \frac{m(v-u)}{t} = \frac{mv - mu}{t}$$

This demonstrates an alternative, more generalized, interpretation of Newton's second law ($F = ma$) in terms of a *change* of momentum, Δp ($= mv - mu$) that occurs in time Δt.

DB

force = rate of change of momentum: $\boldsymbol{F} = \dfrac{\Delta \boldsymbol{p}}{\Delta \boldsymbol{t}}$

This equation allows for the possibility of a changing mass, whereas the use of $F = ma$ assumes a constant mass. An application of this is given later in the section on explosions and propulsion.

Inquiry 2: Collecting and processing data

Interpreting results

Significant figures

An answer should not have more significant figures than the least precise of the data used in the calculation.

The more precise a measurement is, the greater the number of **significant figures** (digits) that can be used to represent it. For example, a mass stated to be 4.20 g (as distinct from 4.19 g or 4.21 g) suggests a greater precision than a mass stated to be 4.2 g.

Significant figures are all the digits used in data to carry meaning, whether they are before or after a decimal point, and this includes zeros.

But sometimes zeros are used without thought or meaning, and this can lead to confusion. For example, if you are told that it is 100 km to the nearest airport, you might be unsure whether it is approximately 100 km, or 'exactly' 100 km. This is a good example of why **scientific notation** is useful. Using 1.00×10^2 km makes it clear that there are 3 significant figures. 1×10^2 km represents much less precision. When making calculations, the result cannot be more precise than the data used to produce it. As a general and simplified rule, when answering questions or processing experimental data, the result should have the same number of significant figures as the data used. If the number of significant figures is not the same for all pieces of data, then the number of significant figures in the answer should be the same as the least precise of the data (which has the fewest significant figures).

You may assume that all the digits seen in the data shown in this book are significant. For example 100 km should be interpreted as three significant figures.

For example, if a mass of 583 g changed velocity by 15 m s^{-1} in two seconds, then the resultant force acting was:

$$F = \frac{\Delta p}{\Delta t} = \frac{(0.583 \times 15)}{2} = 4.3725\,\text{N}$$

(showing all figures seen on calculator display).

But, since the time was only given to one significant figure, then the answer should have the same: $F = 4$ N.

Maybe this can seem unsatisfactory, but remember that when the time is quoted as 2 s, it simply means that it was more than 1.5 s and less than 2.5 s. (A time of 1.5 s would give an answer of 5.8 N, a time of 2.5 s would give an answer of 3.5 N.) If the time had been 2.0 s, then the quoted answer for the force should be 4.4 N. If the time had been 2.00 s, then the quoted answer for the force should still be 4.4 N, because the velocity was only given to 2 significant figures.

◆ **Scientific notation** Every number is expressed in the following form: $a \times 10^b$, where a is a decimal number larger than 1 and less than 10 and b is an exponent (integer).

◆ **Significant figures (digits)** All the digits used in data to carry meaning, whether they are before or after a decimal point.

◆ **Collision** Two (or more) objects coming together and exerting forces on each other for a relatively short time.

◆ **Explosion** Term used in physics to describe when two or more masses, which were initially at rest, are propelled apart from each other.

Conservation of momentum in collisions and explosions

SYLLABUS CONTENT

- Elastic and inelastic collisions of two bodies.
- Energy considerations in elastic collisions, inelastic collisions, and explosions.

We will use the word '**collision**' to describe any event in which two, or more, objects move towards each other and exert forces on each other. In physics this term is not limited to its typical everyday use to describe accidental events, often involving large forces. The term '**explosion**' will be used to describe any event in which internal forces within a stationary system result in separate parts of the system moving apart. Everyday usage of the term is much more dramatic.

■ **Figure A2.61** Newton's cradle is a demonstration of collisions

Collisions

We can *always* use the law of conservation of momentum to help predict what will happen immediately after a collision, but we must also have some other information. For example, the results of two tennis balls colliding will be very different from two cushions colliding and neither can be predicted by using *only* the law of conservation of momentum. Consider the following worked example.

WORKED EXAMPLE A2.10

An object A of mass of 2.1 kg was moving to the left with a velocity of $0.76\,\mathrm{m\,s^{-1}}$. At the same time, object B of mass 1.2 kg was moving in the opposite direction with a velocity of $3.3\,\mathrm{m\,s^{-1}}$. Discuss what happens after the collision.

Momentum of A = $2.1 \times 0.76 = 1.60\,\mathrm{kg\,m\,s^{-1}}$. But this is a magnitude only. We have not considered direction:

If we choose that velocity to the left is positive, momentum of A = $2.1 \times (+\,0.76) = +1.60\,\mathrm{kg\,m\,s^{-1}}$ (to the left).

(Alternatively, if we prefer to say velocity to the right is positive, then we get: momentum of A = $2.1 \times (-0.76) = -1.60\,\mathrm{kg\,m\,s^{-1}}$ (to the left)

Using velocity and momentum to the left to be positive:

momentum of B = $1.2 \times (-3.3) = -3.96\,\mathrm{kg\,m\,s^{-1}}$ (to the right)

The combined momentum of A and B before the collision = $1.60 + (-3.96) = -2.36\,\mathrm{kg\,m\,s^{-1}}$ (to the right)

The law of conservation of momentum tells us that after the collision, this momentum will be the same (assuming there is no resultant external force). But we need further information to determine exactly what happened. That information may come in the form of identifying the type of collision (see below), or telling us what happened to one of the objects, so that we can calculate what happened to the other, as follows:

If, after the collision, object A moved to the left with a velocity of $0.87\,\mathrm{m\,s^{-1}}$, what happened to object B?

After the collision, momentum of A + momentum of B = $-2.36\,\mathrm{kg\,m\,s^{-1}}$

$[2.1 \times (-0.87)] + (1.2 \times v_B) = -2.36\,\mathrm{kg\,m\,s^{-1}}$

$v_B = -0.44\,\mathrm{m\,s^{-1}}$ (to the left)

All of this has been explained in detail to help understanding. More directly it can be represented by: momentum before collision = momentum after collision

$[2.1 \times (+0.76)] + [1.2 \times (-3.3)] = [2.1 \times (-0.87)] + (1.2 \times v_B)$

$v_B = -0.44\,\mathrm{m\,s^{-1}}$ (to the left)

Top tip!

In Topic A.3, we will introduce the law of conservation of energy and the concept of kinetic energy, which is the energy of moving masses, calculated by $E_k = \frac{1}{2}mv^2$. That knowledge is needed in order to understand the rest of this section on collisions.

You may prefer to delay the rest of this topic on collisions and explosions until Topic A.3 has been covered in detail.

Kinetic energy in collisions

We need to consider the transfer of energy in a collision. Any moving object has kinetic energy and during a collision some, or all, of this energy will be transferred between the colliding objects. Typically, some energy, perhaps most of the energy, will be transferred to the surroundings as thermal energy and maybe some sound. We can identify the extreme cases:

◆ **Collisions** In an **elastic collision** the total kinetic energy before and after the collision is the same. In any **inelastic collision** the total kinetic energy is reduced after the collision. If the objects stick together it is described as a **totally inelastic collision**.

◆ **Macroscopic** Can be observed without the need for a microscope.

◆ **Microscopic** Describes anything that is too small to be seen with the unaided eye.

A collision in which the *total* kinetic energy before and after the collision is the same is called an **elastic collision.**

All other collisions can be described as **inelastic collisions**, meaning that kinetic energy has not been conserved. In everyday, **macroscopic** events, elastic collisions are a theoretical ideal and they do not happen perfectly. However, elastic collisions are common for **microscopic** particle collisions.

A collision after which the colliding objects stick together is called a **totally inelastic collision**.

In a totally inelastic collision, the maximum possible amount of kinetic energy is transferred from the moving objects to the environment.

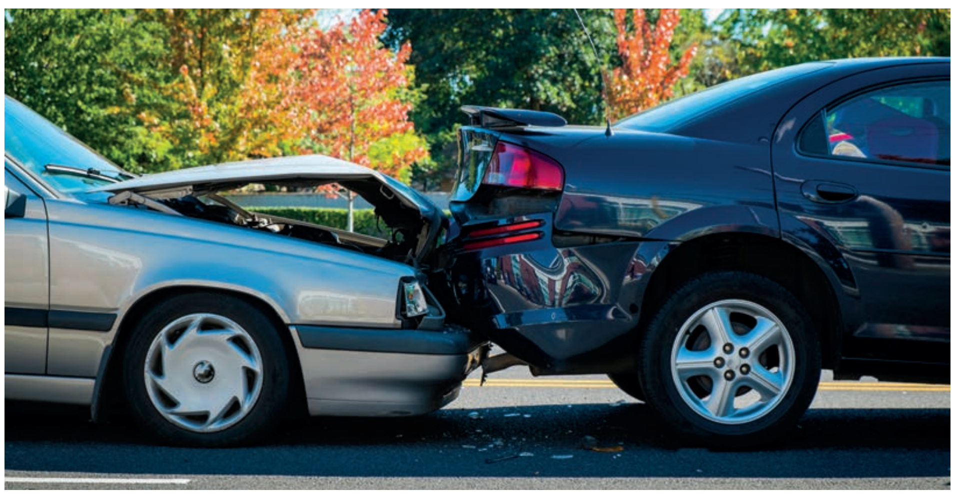


Figure A2.62 An inelastic collision

Nature of science: Models

Macroscopic and microscopic

In general, physicists use the terms:

- **macroscopic** to describe events that can be observed with the unaided eye
- **microscopic** to describe events on the molecular, atomic, or subatomic scale.

It was not until scientists began to realize that matter consisted of atoms and molecules (that could not be seen), that many observations of the world around us could be explained.

Perhaps the best example of an (almost) elastic macroscopic collision is that between steel spheres. If a stationary sphere is struck by an identical moving sphere, the moving sphere stops and the other sphere continues with the velocity of the first. 'Newton's cradle', as seen in Figure A2.61 is a famous demonstration of this.

It is easy to find examples where *all* kinetic energy appears to have been lost, for example, when a student jumps down to the floor. The student has had a totally inelastic collision with the Earth, but the change to the motion of the Earth is insignificant and unobservable.

WORKED EXAMPLE A2.11

A 2.1 kg trolley moving at 0.82 m s^{-1} collides with a 1.7 kg trolley moving at 0.98 m s^{-1} in the opposite direction. After the collision, the 1.7 kg trolley reverses direction and moves at 0.43 m s^{-1}.

a Discuss what happened to the other trolley.

b Without making any calculations, comment on the difference in total kinetic energy before and after the collision.

c If the collision had been totally inelastic, what would have happened after the collision?

Answer

a Total momentum before = total momentum after

$(2.1 \times 0.82) + (1.7 \times -0.98) = (2.1 \times v) + (1.7 \times 0.43)$

Velocities in the original direction have been given a + sign and velocities in the opposite direction are given a − sign (or it could be the other way around).

$0.056 = 2.1v + 0.731$

$v = -0.32$ m s^{-1}. The – sign shows us that the trolley reverses its direction of motion.

b Both velocities have been reduced, so the total kinetic energy is significantly less.

c The trolleys will stick together if the collision is totally inelastic.
Total momentum before = total momentum after

$(2.1 \times 0.82) + (1.7 \times -0.98) = (2.1 + 1.7) \times v$

$0.056 = 3.8v$

$v = +0.015$ m s^{-1}. The + sign shows us that they move in the original direction of the 2.1 kg trolley.

The combined trolleys are moving slowly, so there has been a considerable loss of kinetic energy.

LINKING QUESTION

- In which way is conservation of momentum relevant to the workings of a nuclear power station?

This question links to understandings in Topic E.4

54 An object of mass 4.1 kg travelling to the right with a velocity of 1.9 m s^{-1} has a totally inelastic collision with a stationary object of mass 5.6 kg. Determine how they move immediately after the collision.

55 A bus of mass 4900 kg travelling at 22 m s^{-1} collides with the back of a 1300 kg car travelling at 16 m s^{-1}. If the car is pushed forward with a velocity of 20 m s^{-1}, calculate the velocity of the bus immediately after the collision.

56 In an experiment to find the speed of a 2.40 g bullet, it was fired into a 650 g block of wood at rest on a friction-free surface. If the block (and bullet) moved off with an initial speed of 96.0 cm s^{-1}. Calculate the speed of the bullet.

57 A ball thrown vertically upwards decelerates and its momentum decreases, although the law of conservation of momentum states that total momentum cannot change. Explain this observation.

58 Figure A2.63 shows two trolleys on a friction-free surface joined together by a thin rubber cord under tension. When the trolleys are released, they accelerate towards each other and the cord quickly becomes loose.

a Show that the two trolleys collide at the 20 cm mark.

b Predict what happens after they collide if they stick together.

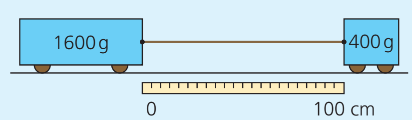


Figure A2.63 Two trolleys on a friction-free surface joined together by a thin rubber cord

59 Two toy cars travel in straight lines towards each other on a friction-free track. Car A has a mass of 432 g and a speed of 83.2 cm s^{-1}. Car B has a mass of 287 g and speed of 68.2 cm s^{-1}. If they stick together after impact, predict their combined velocity.

60 A steel ball of mass 1.2 kg moving at 2.7 m s^{-1} collides head-on with another steel ball of mass 0.54 kg moving in the opposite direction at 3.9 m s^{-1}. The balls bounce off each other, each returning back in the direction it came from on a horizontal surface.

a If the smaller ball had a speed after the collision of 6.0 m s^{-1}, use the law of conservation of momentum to predict the speed of the larger ball.

b In fact, the situation described in part **a** is not possible. Discuss possible explanations of why not.

ATL A2C: Thinking skills

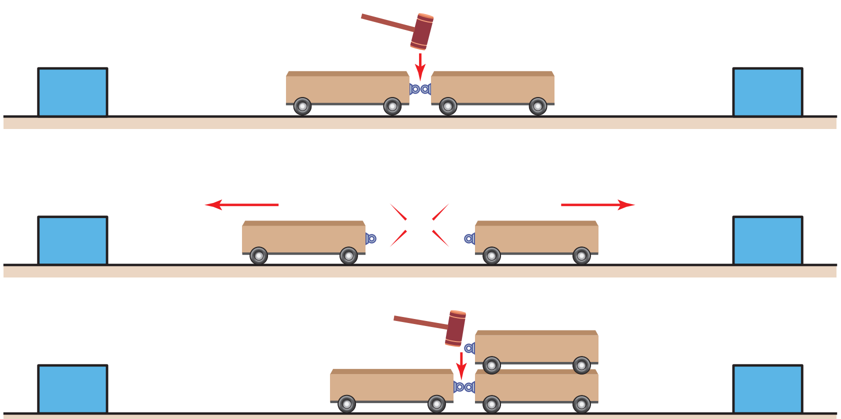
Reflecting on the credibility of results

A student carried out an experiment into the momentum of colliding trolleys on a horizontal runway. A trolley of mass 2.0 kg and speed 80 cm s^{-1} collided with a trolley of mass 1.0 kg and speed 220 cm s^{-1} travelling in the opposite direction. After the collision, both the trolleys reversed their directions and the student measured the speeds of both trolleys to be 60 cm s^{-1}.

Explain why the student must have made a mistake.

Explosions

Figure A2.64 shows a possible laboratory investigation into a one-dimensional 'explosion'. A blow from the hammer releases springs which push the previously stationary trolleys apart. If the trolleys are identical, they will move apart with equal speeds. If the mass on one side is doubled, as shown in the third drawing, the speeds will be in the ratio 2:1, the more massive trolley will move more slowly



■ **Figure A2.64** A simple 'explosion'

■ **Figure A2.65** Firing a cannon

◆ **Recoil** When a bullet is fired from a gun (or similar), the gun must gain equal momentum in the opposite direction.

Firing a gun, or a cannon, is a more dramatic example. See Figure A2.65. Since there is zero momentum to begin with, the momentum of the bullet / cannon ball must be equal and opposite to the momentum of the gun itself (or cannon), so that the total momentum after firing is also zero. The word **recoil** is used to describe this 'backwards' motion.

WORKED EXAMPLE A2.12

A rifle of mass 1.54 kg fires a bullet of mass 22 g at a speed of $250\,\mathrm{m\,s^{-1}}$. Calculate the recoil speed of the rifle.

Answer

Total momentum before = total momentum after

$0 = (1.54 \times v) + (0.022 \times 250)$

$v = -3.6\,\mathrm{m\,s^{-1}}$.

The '–' sign shows us that the gun moves in the opposite direction to the bullet's velocity.

61 In an experiment similar to that shown in the first drawing of Figure A2.64, after being released one trolley with a mass of 980 g recoiled with a velocity of $0.27\,\mathrm{m\,s^{-1}}$. What was the speed of the other trolley, which had a mass of 645 g?

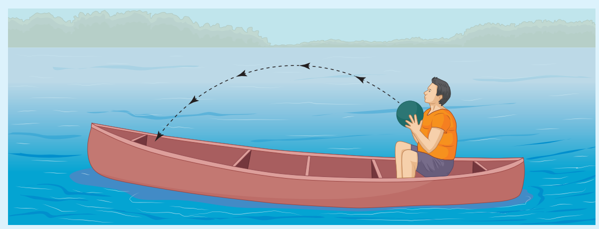
62 An isolated and 'stationary' astronaut of mass 65 kg accidentally pushes a 2.3 kg hammer away from her body with a speed of $80\,\mathrm{cm\,s^{-1}}$.

- **a** Outline a reason why the word 'stationary' has been put in quotation marks.
- **b** Predict what happened to the astronaut.
- **c** Suggest how she can stop moving.

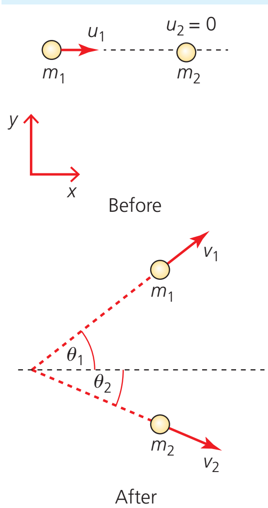
63 Cannons have been used extensively in wars for hundreds of years. A large cannon from 200 years ago could fire a 25 kg cannon ball with a speed of over $150\,\mathrm{km\,h^{-1}}$ ($42\,\mathrm{m\,s^{-1}}$). The recoil momentum of these cannons could be dangerous, although the recoil speed was limited by the very large mass of the cannon. If the speed of recoil was $0.30\,\mathrm{m\,s^{-1}}$, calculate the mass of the cannon.

64 Figure A2.66 shows a heavy ball being thrown from one end of a canoe to the other. Describe what will happen to the canoe (and the passenger) when:

- **a** the ball is being thrown
- **b** the ball is in the air
- **c** the ball lands back in the canoe. Assume that the water does not resist any possible movement of the canoe.



■ **Figure A2.66** A heavy ball being thrown from one end of a canoe to the other



■ **Figure A2.67** Two objects colliding in two dimensions

Collisions and explosions in two dimensions

So far, we have only considered interactions in one direction. This section extends the study to two dimensions. *It is aimed at Higher Level students only.* We will consider that the interacting objects behave as point particles.

Figure A2.67 shows two objects, m_1 and m_2, before and after a collision. In this example m_2 was stationary before the collision.

For collisions or explosions in two dimensions the law of conservation of momentum can be applied in two perpendicular directions.

For the example shown in Figure A2.67 we need to know the components of v_1 and v_2 in the x and y directions:

$$v_{1x} = v_1 \cos\theta_1 \qquad v_{1y} = v_1 \sin\theta_1$$

$$v_{2x} = v_2 \cos\theta_2 \qquad v_{2y} = v_2 \sin\theta_2$$

HL ONLY

WORKED EXAMPLE A2.13

Considering Figure A2.67, let $m_1 = 0.50\,\text{kg}$, $m_2 = 0.30\,\text{kg}$ and $u_1 = 4.0\,\text{m s}^{-1}$.
If the angles were $\theta_1 = 36.9°$ and $\theta_2 = 26.6°$, determine the value of v_2 if v_1 was $2.0\,\text{m s}^{-1}$.

Answer

Applying conservation of momentum in the y-direction:

$0 = m_1 v_1 \sin\theta_1 + m_2 v_2 \sin\theta_2$

$0 = 0.600 + 0.134 v_2$

$v_2 = -4.5\,\text{m s}^{-1}$, the negative sign shows that it is moving in the negative y-direction.

Alternatively, and less simply, we can apply conservation of momentum in the x-direction:

$m_1 u_1 = m_1 v_1 \cos\theta_1 + m_2 v_2 \cos\theta_2$

$(0.50 \times 4.0) = (0.50 \times 2.0 \times \cos 36.9°) + (0.30 \times v_2 \times \cos 26.6°)$

$2.0 = 0.800 + (0.268 \times v_2)$

$v_2 = 4.5\,\text{m s}^{-1}$, as before.

WORKED EXAMPLE A2.14

A stationary small mass of mass 5.0 g explodes into three particles.

One particle, A, of mass 1.5 g moves with a velocity of $23\,\text{m s}^{-1}$ in a direction that we will call the x-direction.

The second particle, B, of mass 2.0 g moves with a velocity of $18\,\text{m s}^{-1}$ in a direction of 60° to the first.

Determine what happened to the third particle, C.

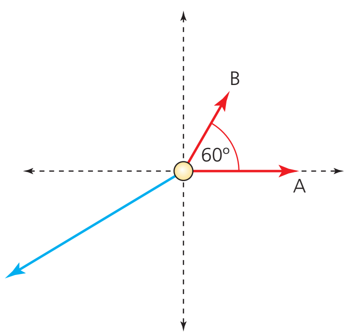


Figure A2.68 A stationary small mass explodes into three particles

Answer

Before the explosion there is zero momentum. After the explosion, in any chosen direction momentum must be conserved: $p_A + p_B + p_C = 0$.

Resolving the velocity of the B particle into two components:

In x-direction, $v_x = 18 \cos 60° = 9.0\,\text{m s}^{-1}$

In y direction, $v_y = 18 \sin 60° = 15.6\,\text{m s}^{-1}$

Consider momentum in the x-direction: $p_A + p_B + p_C = 0$.

$(1.5 \times 23) + (2.0 \times 9.0) = 52.5\,\text{g m s}^{-1} = -p_C$

Consider momentum in the y-direction: $0 + (2.0 \times 15.6) = 31.2\,\text{g m s}^{-1} = -p_C$

Dividing momentum by mass ($m = 5.0 - 1.5 - 2.0 = 1.5$) gives us the two components of the velocity of C: $-35\,\text{m s}^{-1}$ and $-21\,\text{m s}^{-1}$.

These two components can be added (using a scale drawing or trigonometry) to determine the actual velocity of C: $41\,\text{m s}^{-1}$ at an angle of 31° to the $-x$-direction.

65 Masses of 200 g and 500 g are travelling directly towards each other with speeds of $1.2\,m\,s^{-1}$ and $0.30\,m\,s^{-1}$, respectively. After they collide, the speed of the 200 g mass reduces to $0.10\,m\,s^{-1}$ as it continues in a direction at 30° to its original motion.
Determine what happened to the other mass.

66 A mass of mass 1.0 kg moving at $2.0\,m\,s^{-1}$ explodes into three parts. One part, which has a mass of 250 g, has a velocity of $8.5\,m\,s^{-1}$ in the original direction of motion. The second part has a mass of 450 g and moves with a velocity of $5.6\,m\,s^{-1}$ at an angle of 90° to the first part. Show that the third part has a speed of $8.4\,m\,s^{-1}$.

HL ONLY

Propulsion

◆ **Propel** Provide a force for an intended motion.

◆ **Jet engine** An engine that achieves propulsion by emitting a fast-moving stream of gas or liquid in the opposite direction from the intended motion.

◆ **Rocket engine** Similar to a jet engine, but there is no air intake. Instead, an oxidant is carried on the vehicle, together with the fuel.

If the ball shown in Figure A2.66 had been thrown over the end of the canoe, the canoe would keep moving to the right (until resistive forces stopped it). An unusual example perhaps, but this shows us a very useful concept: to start, or maintain motion (**propel**), we can create momentum in the opposite direction. This can be restated using Newton's third law: if we want a force to move us to the right (for example), we exert a force on the surroundings to the left. The person in the boat pushes the ball to the left and the ball pushes the person (and the boat) to the right. Using friction for walking and car movement has already been discussed.

A boat can be pushed forward by pushing water backwards, using an oar, or a propeller. See Figure A2.69. The momentum of the boat forwards is equal and opposite to the momentum of the water backwards.

A propeller can also be used for a small airplane, but typically the propeller needs to be much larger and rotate faster, because the density of air is much less than water. Larger aircraft use the same conservation of momentum principle, but in a different way:

There are many designs of **jet engines**, but the basic principle is that they take in the surrounding air and use it to burn vaporized fuel. The resulting hot exhaust gases are ejected at the back of the engine with considerable momentum (much greater than the momentum of the air input), the difference is equal and opposite to the forward momentum given to the aircraft.

Rocket engines use the same principle, but they travel where there is little or no air, so they use oxygen that has been stored on the vehicle.

■ **Figure A2.69** Boat propeller

■ **Figure A2.70** A Chinese rocket launching a spacecraft to Mars

HL ONLY

WORKED EXAMPLE A2.15

A rocket is ejecting exhaust gases at a rate of $1.5 \times 10^4\,\text{kg}\,\text{s}^{-1}$. If the speed of the exhaust gases (relative to the rocket) is $2.3 \times 10^3\,\text{m}\,\text{s}^{-1}$, what is the forward force acting on the rocket?

Answer

$$F = \frac{\Delta p}{\Delta t} \Rightarrow \left(\frac{\Delta m}{\Delta t}\right) \times v = (1.5 \times 10^4) \times (2.3 \times 10^3) = 3.5 \times 10^7\,\text{N}$$

67 A rocket's mass at lift-off was $2.7 \times 10^6\,\text{kg}$. If gases were ejected at a rate of $1.9 \times 10^4\,\text{kg}\,\text{s}^{-1}$ with a speed of $2.0 \times 10^3\,\text{m}\,\text{s}^{-1}$:

- **a** determine the initial acceleration of the rocket
- **b** explain why the acceleration will increase as the rocket rises, while the engines provide the same force.

Forces acting for short times: impulses

SYLLABUS CONTENT

- ▶ A resultant force applied to a system constitutes an impulse, J, as given by: $J = F\Delta t$, where F is the average resultant force and Δt is the time of contact.
- ▶ The applied external impulse equals the change in momentum of the system.

◆ **Impulse** The product of force and the time for which the force acts.

Many forces only act for a short time, Δt. Clearly the longer the time for which a force acts, the greater its possible effect, so the concept of **impulse**, J, becomes useful:

impulse, $\boldsymbol{J = F\Delta t}$ SI unit: N s

If a force varies during an interaction, we can use an average value to determine the impulse.

We have seen that $F = \dfrac{\Delta p}{\Delta t}$ which can be rearranged to give $F\Delta t = \Delta p$, showing us that

impulse, $J = \Delta p$ (change of momentum)

An impulse on an isolated object results in a change of momentum, which is numerically equal to the impulse. The units N s and $\text{kg}\,\text{m}\,\text{s}^{-1}$ are equivalent to each other.

WORKED EXAMPLE A2.16

A constant force of 12.0 N acts on a stationary mass of 0.620 kg for 0.580 s.

- **a** Calculate the impulse applied to the mass.
- **b** Calculate the change of momentum of the mass.
- **c** Calculate the final velocity of the mass.

Answer

a $J = F\Delta t = 12.0 \times 0.580 = 6.96\,\text{N s}$

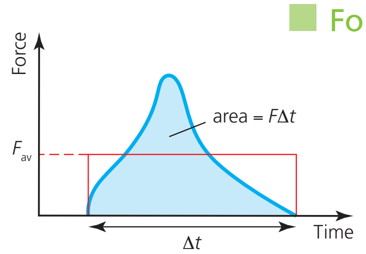
b $6.96\,\text{kg}\,\text{m}\,\text{s}^{-1}$ (or Ns)

c $\Delta p = m\Delta v$

$6.96 = 0.620 \times \Delta v$

$\Delta v = 11.2\,\text{m}\,\text{s}^{-1}$

The final velocity could also be determined by use of $F = ma$ and $v = u + at$.



■ **Figure A2.71** Graph showing how a force varies with time

Force–time graphs

In many simple impulse calculations, we may assume that the forces involved are constant, or that the average force is half of the maximum force. For more accurate work this is not good enough, and we need to know in detail how a force varies during an interaction. Such details are commonly represented by force–time graphs. The curved line in Figure A2.71 shows an example of a force varying over a time Δt.

The area under any force–time graph for an interaction equals force × time, which equals the impulse (change of momentum).

This is true whatever the shape of the graph. The area under the curve in Figure A2.71 can be estimated by drawing a rectangle of the same area (as judged by eye), as shown in red. F_{av} is then the average force during the interaction.

Inquiry 2: Collecting and processing data

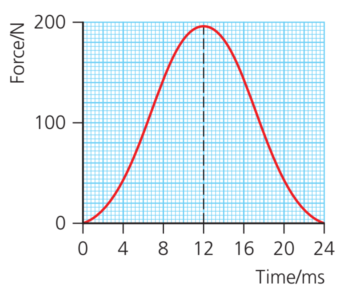
Applying technology to collect data

Force sensors that can measure the magnitude of forces over short intervals of time can be used with data loggers to gather data and draw force–time graphs for a variety of interactions, both inside and outside a laboratory. Stop-motion replay of video recordings of collisions can also be very interesting and instructive.

Force–time graphs can be helpful when analysing any interaction, but especially impacts involved in road accidents and sports.

WORKED EXAMPLE A2.17

Figure A2.72 shows how the force on a 57 g tennis ball moving at 24 m s^{-1} to the right varied when it was struck by a racket moving in the opposite direction.



■ **Figure A2.72** Force–time graph for striking a tennis ball

a Estimate the impulse given to the ball.

b Calculate the velocity of the ball after being struck by the racket.

c The ball is struck with the same force with different rackets. Explain why a racket with looser strings could return the ball with greater speed.

d Suggest a disadvantage of playing tennis with a racket with looser strings.

Answer

a Impulse = area under graph ≈ 200 × (12 × 10^{-3}) = 2.4 Ns to the left

b $m\Delta v = 2.4$

$$\Delta v = \frac{2.4}{0.057} = 42\,\text{m s}^{-1} \text{ to the left}$$

$v_{final} - v_{initial} = -42$ (velocity to the left chosen to be negative)

$v_{final} - (+24) = -42$

$v_{final} = -18$ m s^{-1} (to the left)

c The time of contact with the ball, Δt, will be longer with looser strings, so that the same force will produce a greater impulse (change of momentum).

d There is less control over the direction of the ball.

68 A ball of mass 260 g falls vertically downwards and hits the ground with a speed of $7.3\,m\,s^{-1}$.

- **a** What was its greatest momentum?
- **b** If it rebounded with a speed of $5.5\,m\,s^{-1}$, calculate the change of momentum.
- **c** Determine the impulse on the ground.
- **d** If the duration of the impact was 0.38 s, calculate the average force on the ball during the collision.
- **e** Estimate the maximum force on the ball.

69 A baseball bat hits a ball with an average force of 970 N that acts for 0.0088 s.

- **a** What impulse is given to the ball?
- **b** What is the change of momentum of the ball?
- **c** The ball was hit back in the same direction that it came from. If its speed before being hit was $32\,m\,s^{-1}$, calculate its speed afterwards. (Mass of baseball is 145 g.)

70 Figure A2.73 shows how the force between two colliding cars changed with time. Both cars were driving in the same direction and after the collision they did not stick together.

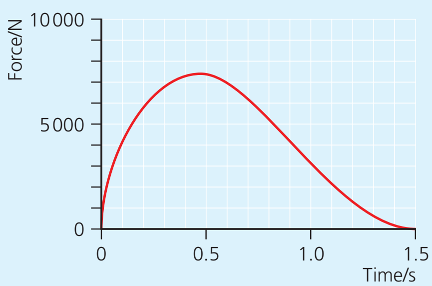


Figure A2.73 How the force between two colliding cars changed with time

- **a** Show that the impulse was approximately 6500 Ns.
- **b** Before the collision the faster car (mass 1200 kg) was travelling at $18\,m\,s^{-1}$. Estimate its speed immediately after the collision.

71 Consider Figure A2.74.

- **a** Discuss how the movement of the karate expert can maximize the force exerted on the boards.
- **b** What features of the boards will help to make this an impressive demonstration?

Figure A2.74 Karate expert

72 A soft ball, A, of mass 500 g is moving to the right with a speed of $3.0\,m\,s^{-1}$ when it collides with another soft ball, B, moving to the left. The time of impact is 0.34 s, after which ball A rebounds with a speed of $2.0\,m\,s^{-1}$.

- **a** What was the change of velocity of ball A?
- **b** What was the change of momentum of ball A?
- **c** Calculate the average force exerted on ball A.
- **d** Sketch a force–time graph for the impact.
- **e** Add to your sketch a possible force–time graph for the collision of hard balls of similar masses and velocities.
- **f** Suggest how a force–time graph for ball B would be different (or the same) as for ball A.

Circular motion and centripetal forces

SYLLABUS CONTENT

- Circular motion is caused by a centripetal force acting perpendicularly to the velocity.
- A centripetal force causes the body to change direction even if the magnitude of its velocity may remain constant.